



Chemistry S Review Guide

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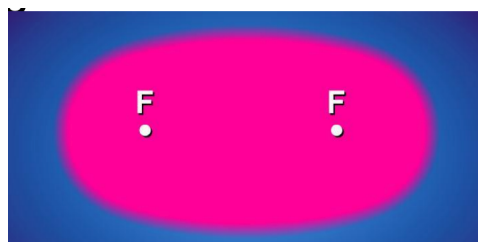
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Chapter 15 Water and Aqueous Systems

15.1 Water and Its Properties

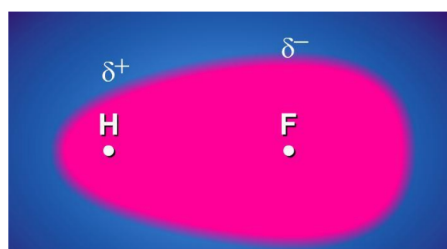
1. Bond polarity

- A **nonpolar bond** is a covalent bond formed when the atoms in the bond pull equally and



the bonding electrons are shared equally.

- A **polar bond** is a covalent bond between atoms in which the electrons are shared unequally.



2. Polarity of molecules

➤ Most molecules with **only nonpolar bonds are nonpolar**. eg: H_2 , C_3H_8

➤ For molecules with polar bonds:

✓ **Bent and pyramidal shape molecules must be polar.**

✓ When the “outside” atoms are identical, molecules with **linear, trigonal planar, tetrahedral shape are nonpolar.**

1. Intermolecular forces

Three types: **dispersion forces, dipole-dipole forces and hydrogen bonds.**

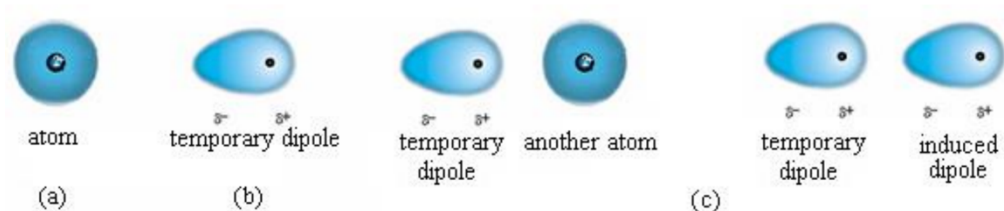
(1) Dispersion Forces

(a) A symmetrical, spherical atom has no separation of charge.

(b) An atom can lose its spherical shape and become a temporary dipole, which has a

separation of opposite charge.

(c) A temporary dipole creates an induced dipole in the adjacent atoms.



Induced Dipole: a dipole that is created by the presence of a neighboring dipole. **Dispersion Force:** a force of attraction between induced dipoles.

-The boiling point of a liquid is a measure of the strength of the attractive forces between molecules in the liquid state. **The stronger the attractive forces, the higher boiling point.**

-As the **size of atoms** increase, boiling point increases.

-Dispersion forces increase with increasing **molecular/atomic size and mass**

(2) Dipole-dipole forces

a. Polar molecules have permanent dipoles.

b. dipole-dipole forces: attraction between opposite charges of neighboring permanent dipoles.

(3) Hydrogen Bonding

-In a covalent bond $H-X$ (X is N, O or F), the partial charges (δ^+ and δ^-) are particularly large due to the large difference in electronegativities between H and X.

4. Water molecules

-**high** surface tension, **low** vapor pressure, and **high** boiling point—result from hydrogen bonding

5. Surface Tension

-The inward force, or pull, that tends to minimize the surface area of a liquid is called surface tension.

- A **surfactant** is any substance that interferes with the hydrogen bonding between water molecules and thereby reduces surface tension.

- Soaps and detergents are surfactants

6. Vapor Pressure

Hydrogen bonding between water molecules also explains water's unusually low vapor pressure.

- An extensive network of hydrogen bonds holds the molecules in liquid water to one another.
- These hydrogen bonds must be broken before water changes from the liquid to the vapor state, so the tendency of these molecules to escape is low and evaporation is slow.

7. Density

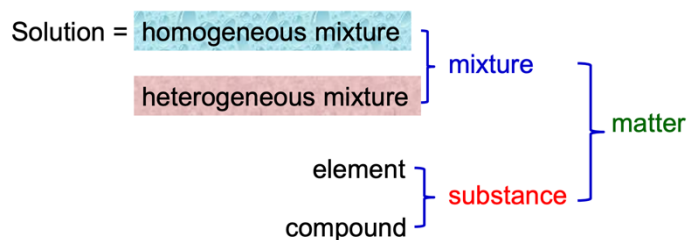
-ice less dense than liquid water.

-Ice has a hexagonal structure. Extensive hydrogen bonding in ice holds the water molecules farther apart in a more ordered arrangement than in liquid water.

15.2 Homogeneous Aqueous Systems

I. Solutions

- aqueous solution: a solution with water as the solvent.



-Solute is normally:

- 1) the component whose original state is different from the state of solution
- 2) or the one that present in small amount.

“Like dissolves like”

Substances with similar **bonding pattern, polarity, or intermolecular attractive forces** tend to be soluble in one another.

- **Nonpolar** substances are more likely to be soluble in **nonpolar** solvents.
- **Ionic and polar solutes** are more likely to be soluble in **polar** solvents.
- **Network solids** are not soluble in either polar or nonpolar solvents.

Solvation: the process of surrounding solute particles with solvent particles to form a solution.

Electrolytes and Nonelectrolytes

electrolyte: a compound that conducts an electric current when it is in an aqueous solution or in the molten state.

e.g., all ionic compounds, some molecular compounds (most acids or bases, water), etc.

nonelectrolyte: a compound that does not conduct an electric current in either an aqueous solution or the molten state.

– Some polar molecular compounds are nonelectrolytes in the pure state but become electrolytes when they dissolve in water.

E.g., ammonia (NH₃ (g)) is not an electrolyte in the pure state.



Hydrates

Hydrate: a compound that contains H₂O units within its crystal structure.

water of hydration: The water contained in a crystal.
(*water of crystallization*)

eg: copper(II) sulfate pentahydrate, **CuSO₄·5H₂O**

-the water is easily lost and regained.

To determine **the percent by mass of water in a hydrate**:

1. Determine the mass of water in one mole of hydrate.
2. Determine **the molar mass of the hydrate**.
3. Calculate using the following equation:

$$\text{Percent by mass H}_2\text{O} = \frac{\text{mass of water}}{\text{mass of hydrate}} \times 100\%$$

Some Common Hydrates		
Formula	Chemical name	Common name
MgSO ₄ ·7H ₂ O	Magnesium sulfate heptahydrate	Epsom salt
Ba(OH) ₂ ·8H ₂ O	Barium hydroxide octahydrate	
CaCl ₂ ·2H ₂ O	Calcium chloride dihydrate	
CuSO₄·5H₂O	Copper(II) sulfate pentahydrate	Blue vitriol
Na ₂ SO ₄ ·10H ₂ O	Sodium sulfate decahydrate	Glauber' s salt
KAl(SO ₄) ₂ ·12H ₂ O	Potassium aluminum sulfate dodecahydrate	Alum
Na ₂ B ₄ O ₇ ·10H ₂ O	Sodium tetraborate decahydrate	Borax

FeSO ₄ ·7H ₂ O	Iron(II) sulfate heptahydrate	Green vitriol
H ₂ SO ₄ ·H ₂ O	Sulfuric acid hydrate (mp 8.6°C)	

-the chart is required. **Needs to be memorized!!**

effloresce(风化): the process in which a hydrate loses its water of hydration.

-Hydrated ionic compounds that have low vapor pressure remove water from moist air to form higher hydrates.

hygroscopic hydrates: Hydrates and other compounds that remove moisture from air.

desiccant: a substance used to absorb moisture from the air and create a dry atmosphere.

Deliquescent (易潮解的): Compounds Some compounds are so hygroscopic that they remove sufficient water from the air to dissolve completely and form solutions.

$$\text{percent by mass H}_2\text{O} = \frac{\text{mass of water}}{\text{mass of hydrate}} \times 100\%$$

15.3 Heterogeneous Aqueous Systems

1. Suspensions

Definition: A suspension is a mixture from which particles settle out upon standing.

Key Characteristics:

Particle Size: $>1000\text{ nm}$

Type of Mixture: Heterogeneous (two substances can be clearly identified).

Filtration: Particles can be separated by filtration.

Example:

A mixture of sand and water is a suspension because sand particles are large enough to be separated by filtration.

Suspension vs. Solution	Suspension	Solution
Particle Size	$>1000\text{ nm}$	$\approx 1\text{ nm}$
Mixture Type	Heterogeneous	Homogeneous
Separation by Filtration	Can be separated	Cannot be separated

2. Colloids

Definition: A heterogeneous mixture containing particles ranging from 1 nm to 1000 nm .

Dispersion Medium: Solid, liquid, or gas.

Key Characteristics:

Intermediate-sized particles that cannot be filtered.

Particles do not settle over time.

Tyndall Effect:

The scattering of visible light by colloidal particles.

Example: A beam of light passing through fog or mist.

Brownian Motion:

The chaotic movement of colloidal particles caused by collisions with molecules in the dispersion medium.

Coagulation:

Colloidal particles adsorb ions from the medium, gaining charge, which prevents aggregation.

Adding electrolytes neutralizes the charge, causing particles to clump together and settle out.

Examples of Colloidal Systems:

System Type	Dispersed Phase	Dispersion Medium	Examples
Gas in Liquid	Gas	Liquid	Whipped cream
Gas in Solid	Gas	Solid	Marshmallow
Liquid in Liquid	Liquid	Liquid	Milk, mayonnaise
Liquid in Gas	Liquid	Gas	Fog, aerosol
Solid in Gas	Solid	Gas	Smoke, dust in air
Solid in Liquid	Solid	Liquid	Paint, blood, starch in water

3. Emulsions

Definition: A colloidal dispersion of a liquid in a liquid.

Emulsifying Agent: Essential for formation and stability of emulsions.

Example: Soap or detergent.

Soap molecules have a polar end (attracted to water) and a nonpolar end (attracted to oil or grease).

Examples of Emulsions: Oil and water with soap, Mayonnaise.

4. Properties of Solutions, Colloids, and Suspensions

Property	Solution	Colloid	Suspension
Particle Type	Ions, atoms, small molecules	Large molecules or particles	Large particles or aggregates
Particle Size	0.1–1 nm	1–1000 nm	1000 nm and larger
Effect of	No scattering	Exhibits Tyndall effect	Exhibits Tyndall effect
Effect of Gravity	Stable, does not separate	Stable, does not separate	Unstable, sediment forms
Filtration	Particles not retained on filter	Particles not retained on filter	Particles retained on filter
Uniformity	Homogeneous	Heterogeneous	Heterogeneous

Chapter 16 Solutions

16.1 Properties of Solutions

1. Solution Formation

3 Factors That Affect How Fast a Substance Dissolves

Agitation (Stirring): Dissolving occurs at the surface of solute particles.

Stirring brings solvent (e.g., water) in contact with the solute's surface, speeding up the dissolving process.

Note: Agitation does not influence the amount of solute that will dissolve.

Temperature: Higher temperatures increase the kinetic energy of water molecules, causing them to move faster.

This increases the frequency of collisions between solvent molecules and solute particles.

Particle Size of the Solute: Smaller particles (e.g., granulated sugar) have a greater surface area exposed to the solvent, increasing the rate of dissolving.

2. Solubility

-Saturated Solution: Contains the maximum amount of solute for a given quantity of solvent at a constant temperature and pressure.

-Dynamic equilibrium exists between the dissolved solute and any undissolved solute.

Solubility is expressed as grams of solute per 100 g of solvent (g/100 g H₂O).

-Unsaturated Solution: Contains less solute than a saturated solution. Additional solute can dissolve until saturation is reached.

Miscible Liquids: Liquids that dissolve in each other in all proportions (e.g.,

water and ethanol).

Immiscible Liquids: Liquids that do not dissolve in each other (e.g., oil and water).

3. Factors Affecting Solubility

(1) Temperature

The solubility of solid, liquid, and gaseous solutes is affected by temperature:

Solids: Solubility generally increases with temperature.

Gases: Solubility decreases with increasing temperature (e.g., carbonated beverages go flat more quickly at room temperature).

(2) Pressure (Henry's Law)

Henry's Law

- A given temperature, the solubility (S) of a gas in a liquid is directly proportional to the pressure (P) of the gas above the liquid

$$\frac{S_1}{P_1} = \frac{S_2}{P_2}$$

Higher pressure: Increases gas solubility.

Lower pressure: Decreases gas solubility.

4. Supersaturated Solutions

-A supersaturated solution contains more solute than it can theoretically hold at a given temperature.

Crystallization can be triggered by adding a small "seed crystal" of the solute.

The rate at which excess solute deposits upon the surface of a seed crystal can be very rapid.



The solution is clear before a seed crystal is added.

Crystals begin to form immediately after the addition of a seed crystal.

Excess solute crystallizes rapidly.

16.2 Concentrations of Solutions

1. Molarity

-The concentration of a solution is a measure of the amount of solute dissolved in a given quantity of solvent.

Dilute solution: Contains a relatively small amount of solute.

Concentrated solution: Contains a large amount of solute.

Molarity (M): The number of moles of solute dissolved in one liter of solution.

-Also known as molar concentration.

$$\text{Molarity (M)} = \frac{\text{moles of solute}}{\text{liters of solution}}$$

2. Making Dilutions

Definition: Reducing the number of moles of solute per unit volume by adding solvent, while the total number of moles of solute remains unchanged.

$$\text{Moles of solute} = M_1 \times V_1 = M_2 \times V_2$$

- M_1 and V_1 are the molarity and the volume of the initial solution.
- M_2 and V_2 are the molarity and volume of the diluted solution.

3. Percent Solutions

Percent by volume of a solution is the ratio of the volume of solute to the volume of solution.

$$\text{Percent by volume (\% (v/v))} = \frac{\text{volume of solute}}{\text{volume of solution}} \times 100\%$$

Percent by mass of a solution is the ratio of the mass of the solute to the mass of the solution.

$$\text{Percent by mass (\% (m/m))} = \frac{\text{mass of solute}}{\text{mass of solution}} \times 100\%$$

Summary: Three Ways to Calculate Solution Concentration

Molarity (M): Moles of solute per liter of solution.

Percent by Volume (%(v/v)): Volume of solute per volume of solution $\times 100\%$.

Percent by Mass (%(m/m)): Mass of solute per mass of solution $\times 100\%$.

16.3 Colligative Properties of Solutions

1. Describing Colligative Properties

Colligative Properties: Properties of solutions that depend only on the number of solute particles, not their identity.

Three important colligative properties:

-Vapor-pressure lowering

-Freezing-point depression

-Boiling-point elevation

Effect of solutes on vapor pressure:

Solute particles reduce the number of solvent particles able to escape into vapor.

Equilibrium is established at a lower vapor pressure.

Ionic vs. Molecular Solutes

Ionic solutes that dissociate have a greater effect on vapor pressure than nondissociating

solutes.

Example effects of 3 moles of solute in water:

NaCl: 6 moles of particles (dissociates into 2 ions).

CaCl₂: 9 moles of particles (dissociates into 3 ions).

Glucose: 3 moles of particles (does not dissociate).

The decrease in vapor pressure is proportional to the number of solute particles.

2. Freezing-Point Depression

When a substance freezes, its particles form an orderly pattern.

Effect of solutes on freezing point:

- Solute disrupts the formation of the solid pattern.
- More kinetic energy must be withdrawn for the solution to solidify.
- The freezing point of a solution is lower than the freezing point of the pure solvent.

Magnitude of freezing-point depression is **proportional** to the number of solute particles dissolved, not their identity.

3. Boiling-Point Elevation

Boiling point: The temperature at which the vapor pressure of a liquid equals atmospheric pressure.

Effect of solutes on boiling point:

- Adding a nonvolatile solute decreases the solvent's vapor pressure.
- More kinetic energy is required to raise the vapor pressure to atmospheric pressure, increasing the boiling point.
- The boiling point of a solution is higher than the boiling point of the pure solvent.
- still proportional

16.4 Calculations Involving Colligative Properties

1. Molality and Mole Fraction

Molality is the number of moles of solute dissolved in 1 kilogram (1000 grams) of solvent.

$$\text{Molality (} m \text{)} = \frac{\text{moles of solute}}{\text{kilogram of solvent}}$$

also known as molal concentration.

Mole Fraction (X) of a solute in a solution is the ratio of moles of the solute to the total number of moles of solvent and solute.

$$X_A = \frac{n_A}{n_A + n_B} \quad X_B = \frac{n_B}{n_A + n_B}$$

Where:

X_A = mole fraction of solute A, X_B = mole fraction of solvent B

n_A = moles of solute, n_B = moles of solvent

2. Freezing-Point Depression and Boiling-Point Elevation

Freezing-Point Depression (ΔT_f)

Freezing-point depression is the difference between the freezing point of the solution and the freezing point of the pure solvent.

Formula:

$$\Delta T_f = K_f \times m$$

K_f = molal freezing-point depression constant

m = molality

Boiling-Point Elevation (ΔT_b)

Boiling-point elevation is the difference between the boiling point of the solution and the boiling point of the pure solvent.

Formula:

$$\Delta T_b = K_b \times m$$

K_b = molal boiling-point elevation constant

m = molality

Constants (K_f and K_b)

Solvent	K_f ($^{\circ}\text{C}/m$)	K_b ($^{\circ}\text{C}/m$)
Acetic Acid	3.90	3.07

Solvent	K _f (°C/ m)	K _b (°C/ m)
Benzene	5.12	2.53
Camphor	37.7	5.95
Cyclohexane	20.2	2.79
Ethanol	1.99	1.19
Nitrobenzene	7.00	5.24
Phenol	7.40	3.56
Water	1.86	0.512

$$100.00^{\circ}\text{C} + 1.54^{\circ}\text{C} = 101.54^{\circ}\text{C}$$

3. Solution Concentration Units

Solution concentration can be expressed as:

Molarity (M): moles of solute per liter of solution.

Molality(m): moles of solute per kilogram of solvent.

Percent by Volume: (volume of solute / volume of solution) $\times$ 100%.

Percent by Mass: (mass of solute / mass of solution) $\times$ 100%.